Arithmetic Sequences and Series

Name _____

Algebra 2 Lesson 1-4 (fast track) Date _____ Period _____

Learning goal: I can write an arithmetic sequence with a recursive definition and with an explicit definition, and translate from one to the other.

Language goal: I can explain how a recursive definition and an explicit definition describe the same sequence, using the words first term, common difference, previous term, and term number.

On the quiz: You will (1) pick the arithmetic sequence that matches a description (bonus); (2) add the lengths of stair tops (full Savvas quiz only); (3) pick the explicit definition for a word problem (Example 3A); (4) fill in the recursive definition for a word problem (Example 3B); (5) add up birthday deposits (full Savvas quiz only). For (3) and (4), find the **first term** and the **common difference** first.

On the Topic 1 test: Question 8 asks you to write the explicit formula from a recursive one and use it (Example 3). Questions 7 (a recursive formula from a word problem, Examples 2 and 4), 9 (the sum of a series, Example 7) and 10 (a reasonable domain, Example 1) come from Examples this packet skips.

arithmetic sequence (secuencia aritmética)	constant difference between consecutive terms: the **common difference** d (diferencia común). $3, 8, 13, 18, \ldots$ has $d = 5$.
recursive definition (definición recursiva)	each term is defined by operations on the **previous term**: a_1 is given, then $a_n = a_{n-1} + d$ for $n > 1$
explicit definition (definición explícita)	finds any term from its **term number** n without the previous term: $a_n = a_1 + d(n-1)$

Example 3: Translate Between Recursive and Explicit Forms

A. Given the recursive definition $a_n = \begin{cases} 3, & n = 1 \\ a_{n-1} + 0.5, & n > 1 \end{cases}$, what is an explicit definition for the sequence?

Use the recursive definition to find a pattern:

$a_1 = 3$

$a_2 = a_1 + 0.5 = 3 + 0.5$

$a_3 = a_2 + 0.5 = [3 + 0.5] + 0.5 = 3 + 2(0.5)$

$a_4 = a_3 + 0.5 = [3 + 2(0.5)] + 0.5 = 3 + 3(0.5)$

Term n has $(n-1)$ common differences added to a_1.

term no. n	**built from** a_1	**term**
1	$3 + 0(0.5)$	3
2	$3 + 1(0.5)$	3.5
3	$3 + 2(0.5)$	4
4	$3 + 3(0.5)$	4.5
n	$3 + (n-1)(0.5)$	

Explicit definition: $\boxed{a_n = 3 + (n-1)(0.5)}$. Check: $a_4 = 3 + 3(0.5) = 4.5$ ✓

B. Given the explicit definition $a_n = 16 - 3(n-1)$, what is the recursive definition for the arithmetic sequence?

1. Match $a_n = a_1 + d(n-1)$: the number multiplying $(n-1)$ is the common difference, $d = -3$.
2. First term: $a_1 = 16 - 3(1-1) = 16$.
3. Check: $a_2 = 16 - 3(2-1) = 13$, and $16 - 3 = 13$ ✓.

Recursive definition:

$$a_n = \begin{cases} 16, & n = 1 \\ a_{n-1} - 3, & n > 1 \end{cases}$$

Say it: In Part B, the first term is _____ and the common difference is _____. The recursive definition finds each term from the _____; the explicit one finds any term from its _____.

Try It 3: Translate between the forms

a. For the recursive definition

$$a_n = \begin{cases} 45, & n = 1 \\ a_{n-1} - 2, & n > 1 \end{cases},$$

what is the explicit definition?

$a_1 =$ _____ $\quad d =$ _____

_____

_____

_____

$a_n =$ _____ + _____$(n-1)$

b. For the explicit definition

$a_n = 1 + 7(n-1)$,

what is the recursive definition?

$a_1 =$ _____ $\quad d =$ _____

_____

_____

_____

$$a_n = \begin{cases} _____, & n = 1 \\ _____, & n > 1 \end{cases}$$

In part b, the common difference is the number multiplying _____, so I found it without listing any terms.

Practice: Word problems: first term and common difference first

a. A tank holds 60 gallons of water. A pump removes 5 gallons per minute. Let a_n be the number of gallons left in the tank after n minutes.

$a_1 =$ _____ (gallons left after 1 minute)

$d =$ _____

Explicit: $a_n =$ _____

Recursive: $a_n = \begin{cases} _____, & n = 1 \\ _____, & n > 1 \end{cases}$

After how many minutes is the tank empty? Show the equation you solve.

_____

_____

A reasonable domain for this sequence:
the integers _____ through _____.

b. In week 1 Marco puts \$15 in a savings jar. Every week after that, he puts in \$5 more than he put in the week before. Let $f(n)$ be the amount he puts in during week n.

Complete the recursive definition:

$$f(n) = \begin{cases} _____, & n = 1 \\ f(n-1) + _____, & n > 1 \end{cases}$$

Explicit: $f(n) =$ _____

Amount in week 10: _____

_____

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
C8R ★	Find terms of an arithmetic sequence	Quiz 1 (bonus): find the common difference
H82 ★	Write a formula for an arithmetic sequence	Quiz 3: the explicit definition
ZAH ★	Write a recursive formula for a sequence	Quiz 4: the recursive definition
W6A	Find the sum of an arithmetic series (optional)	Quiz 2 and 5 (full Savvas quiz only); Topic 1 Assessment sums

Exit question: Who is correct?

Felipe and Gregory are given the arithmetic sequence $-1, 6, 13, \ldots$. Gregory wrote the explicit definition $a_n = -1 + 7(n-1)$ for the sequence. Felipe wrote the definition as $a_n = 7n - 8$. Which one of them is correct? Explain.

The first term is _____ and the common difference is _____, so $a_n =$ _____ + _____$(n-1)$.

Simplify Gregory's definition: $-1 + 7(n-1) =$ _____

Check term number $n = 3$ in both definitions:

Who is correct: _____, because

Before the quiz, finish these sentences.

To write an explicit definition, I find the _____ and the _____.

A recursive definition finds each term from the _____ term.

When the terms go down by the same amount each time, the common difference is _____ (positive or negative).